# LabAssignment-10.2

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# Lab 10 – Code Review and Quality: Using AI to Improve Code Quality and Readability

## Lab Objectives

• Use AI for automated code review and quality enhancement.
• Identify and fix syntax, logical, performance, and security issues in Python code.
• Improve readability and maintainability through structured
Week5 - Tuesday
refactoring and
comments.
• Apply prompt engineering for targeted improvements.
• Evaluate AI-generated suggestions against PEP 8 standards and software engineering best practices

## Task 1 – Error Detection and Correction Issues in Given Code

- Missing colon : after function definition.
- Wrong indentation.
- Missing colon in for loop.
- Variable total not defined (should return sum).
- Avoid using built-in name sum as a variable.

### Corrected Code

[1]
✓ 0s

```python
def calculate_total(nums):
    total = 0
    for n in nums:
        total += n
    return total

# Example usage
numbers = [1, 2, 3, 4]
print(calculate_total(numbers))
```

```
10
```

## Fix Summary

- Added missing : symbols.
- Corrected indentation.
- Replaced undefined variable.

- Used total instead of built-in sum.

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## Task 2 – Code Style Standardization (PEP 8)

### Refactored Code

[2]
0s

```python
def find_sum(a, b):
    """Return the sum of two numbers."""
    return a + b


result = find_sum(5, 10)
print(result)
```

```
15
```

### Improvements

- Function name changed to snake_case.
- Added docstring.
- Proper spacing and formatting.

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## Task 3 – Code Clarity Improvement

### Refactored Code

[3]
0s

```python
def calculate_adjusted_value(number, multiplier):
    """Subtract twice the multiplier from the number."""
    return number - (multiplier * 2)


result = calculate_adjusted_value(10, 3)
print(result)
```

```
4
```

### Improvements

- Meaningful function and variable names.
- Added explanation through docstring.
- Improved readability.

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## Task 4 – Structural Refactoring (Reusable Functions)

### Refactored Code

```python
def greet(name):
    """Print a greeting message for the given name."""
    print(f"Hello {name}")


names = ["Ram", "Sita", "Ravi"]

for name in names:
    greet(name)
```

```
Hello Ram
Hello Sita
Hello Ravi
```

### Improvements

- Removed repetition.
- Created reusable function.
- Used loop for scalability.

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## Task 5 – Efficiency Enhancement

### Optimized Code

```python
# Using list comprehension (faster than loop + append)
numbers = [i * i for i in range(1, 500000)]

print(len(numbers))
```

```
499999
```

### Why Faster?

- List comprehensions are optimized in Python.
- Avoids repeated method calls (append).
- Cleaner and more memory-efficient.

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## Conclusion (For Lab Submission)

Using AI-assisted review helped to:

- Fix syntax and logical errors.
- Improve code readability and structure.
- Ensure compliance with **PEP 8**.
- Reduce redundancy through modular design.
- Optimize performance using Python best practices.